

Name: V. Saatwik
SRN: PES1UG24CS509
Section: I

USE CASE FLOW DOCUMENT

USE CASE: Use Case: Reserve Charging Slot (UC-03)

Primary Actor: EV Driver Secondary Actor: Payment Gateway

Preconditions:

- The driver is logged in, and their vehicle connector type is known to the system.
- The system has current real-time availability data for the selected station.

Postconditions (success):

- A confirmed reservation exists for the selected socket and time slot.
- The socket is locked and hidden from other drivers for the reserved window.

Main Success Scenario:

1. The driver selects a charging station from the search results (UC-01).
2. System displays available sockets compatible with the driver's connector type.
3. The driver selects a socket and a 30-minute time slot.
4. System checks that the slot does not overlap with an existing reservation.
5. System temporarily locks the socket for that time slot.
6. System invokes Process Payment (UC-06) to authorize a hold on the driver's payment method.
7. Payment Gateway confirms authorization.
8. The system confirms the reservation and sends a confirmation notification to the driver.

Alternate Flow — 6a. Payment Declined:

- 6a1. Payment Gateway returns a declined authorization.
- 6a2. System releases the temporary socket lock immediately.
- 6a3. System displays a "payment failed" message and prompts the driver to try another payment method.
- 6a4. If the driver retries successfully, flow resumes at step 6. If the driver cancels or fails twice, the reservation attempt is cancelled, and the socket returns to available status.